

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

IT-843 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Operation Research
 (Elective-IV)
 Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. Use two phase method to solve the following L.P.P.:

$$\begin{aligned} \text{Min } Z &= -3x_1 + x_2 \\ \text{s.t. } 2x_1 + x_2 &\geq 2 \\ x_1 + 3x_2 &\leq 2 \\ x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

2. A paper mill produces two grades of paper namely X and Y. Owing to raw material restrictions, it cannot produce more than 400 tons of grade X and 300 tons of grade Y in a week. There are 160 production hours in a week. It requires 0.2 and 0.6 hours to produce a ton of products X and Y, respectively with corresponding profits of ₹200 and ₹500 per ton formulate the above as an LPP to maximize profit and find the optimum product mix.

3. Solve the following transportation problem and find the optimal transportation cost:

	x	y	z	Supply
A	8	9	6	18
B	6	11	5	20
C	3	8	7	18
Demand	15	16	12	

4. Solve the following Integer Programming Problem

$$\begin{aligned} \text{Maximize } Z &= 7x_1 + 9x_2 \\ \text{subject to: } -x_1 + 3x_2 &\leq 6 \\ 7x_1 + x_2 &\leq 35 \\ 0 &\leq x_1 \\ x_2 &\leq 7 \end{aligned}$$

and $x_1, x_2 \geq 0$ and are Integers

[2]

5. A company has a demand of 12000 units/year for an item and it can produce 2000 such items per month. The cost of one set up is Rs. 400 and the holding cost / unit / month is Rs. 0.15. Find the optimum lot size and the total cost per year, assuming the cost of one unit as Rs. 4.00. Also find the maximum inventory and total time.

OR

Explain the basic queuing process. What are the important random variates in a queuing system to be investigated?

6. Explain the terms:
- i) Maximin criteria
 - ii) Optimality Strategy
 - iii) Saddle point
7. Define the following inventory cost:
- i) Item cost
 - ii) Set-up cost
 - iii) Holding cost
 - iv) Stock-out cost
8. Answer any four of the following:
- a) Write the disadvantages of the Big-M Simplex method.
 - b) Explain the cutting plane method of solving an integer programming problem.
 - c) Explain with examples the failure mechanism of items.
 - d) Write a short note on Inventory control.
 - e) Write the postulates for Poisson Process in Queuing Theory.
 - f) Write a short note on M/M/1 models.
